

L04 – Unbalanced (Three-phase) Power Flow

Foundations Lecture Series

Course Instructor

Microgrid 3-phase Security AI Project

Foundations Lecture 4

Keywords: symmetrical components, sequence networks, VUF, runpp_3ph

Lecture roadmap

- 1 Sources of unbalance in real networks
- 2 Symmetrical components: positive, negative, zero
- 3 Sequence networks and the Fortescue transformation
- 4 Three-phase power-flow formulation
- 5 Voltage Unbalance Factor (VUF)
- 6 `pp.runpp_3ph` in practice
- 7 Why microgrid security analysis needs this lecture

Prerequisite

L01 (PF basics), L03 (balanced assumption). L02 helpful for distribution context.

Where does unbalance come from?

TODO: *Bulleted list with visual examples: single-phase loads, single-phase rooftop PV, EV chargers on one phase, asymmetric line geometry, untransposed lines, single-phase laterals on LV feeders.*

What unbalance does to operations

TODO: *Per-phase undervoltage even when average voltage looks fine. Per-phase overload even when balanced loading is below 100%. Neutral currents. Increased losses. Motor heating. Connect to the project's violation definitions.*

The Fortescue transformation

TODO: *Decompose any three-phase phasor set into positive-, negative-, and zero-sequence components. Define $a = e^{j2\pi/3}$ and the transformation matrix A . Inverse transformation. Visual: three rotating phasor sets.*

Sequence networks

TODO: *Show how the three-phase network decouples into three independent sequence networks under linear, symmetric conditions. Mention that grounding configuration affects the zero-sequence network specifically.*

Phase-domain vs sequence-domain formulations

TODO: *Explain that you can either solve three coupled phase equations directly or three independent sequence equations after transformation. pandapower's `runpp_3ph` uses sequence frame: positive via NR, negative and zero via current injection.*

Asymmetric loads and sources

TODO: *Walk through how a single-phase load is modeled in pandapower with `create_asymmetric_load`, including the choice of phase A/B/C. Mirror discussion for `create_asymmetric_sgen`. Note the wye/delta connection options.*

Defining VUF

TODO: *The IEEE/IEC definition: $VUF = |V_{neg}|/|V_{pos}|$. A simpler engineering proxy: max phase voltage deviation divided by average. State the typical regulatory limit ($\sim 2\%$).*

VUF as a violation in our project

TODO: *Forward-reference: VUF is one of the five violation types in the project's labeling (L01 of weekly material). Even when no phase is under/over voltage individually, VUF can exceed 2% and that itself is a violation.*

Minimal three-phase example

TODO: *Code: build a 3-bus unbalanced toy system in pandapower, add asymmetric loads on different phases, call `pp.runpp_3ph(net)`, show `res_bus_3ph` and `res_line_3ph` columns. Compare to balanced `runpp` on the same system.*

Reading per-phase results

TODO: Walk through key columns: *vm_a_pu*, *vm_b_pu*, *vm_c_pu*, *unbalance_percent*, *per-phase loadings*, *neutral current*.

What pandapower 3.x does (and does not) support

TODO: Quick note: *runpp_3ph* solves the sequence-frame PF for grid-connected radial / weakly-meshed networks. It does not model islanded grid-forming inverters or droop control. State as a known limitation that the project also inherits.

Connecting to the project

TODO: *Recap: a microgrid is LV + single-phase loads + single-phase DER + possibly islanded + needs N-1 security analysis. Every reason listed in slide 1 of L04 is present. Hence the entire project is built on `runpp_3ph`.*

Homework

TODO: *Suggest: take a balanced 5-bus feeder, add a single-phase 2 kW load on phase A only, run both `runpp` and `runpp_3ph`, compare per-phase voltages. Compute VUF by hand from V_{pos} , V_{neg} .*

Exit checklist

TODO: *Items: (1) state Fortescue, (2) recognize when a problem needs $runpp_3ph$, (3) compute VUF, (4) read res_bus_3ph , (5) explain the five violation types in the project's labeling.*

TODO: Fortescue 1918, Grainger & Stevenson on symmetrical components, Kersting on three-phase distribution, pandapower "Asymmetric / Three-phase Power Flow" docs, IEEE Std 1159 on power quality.